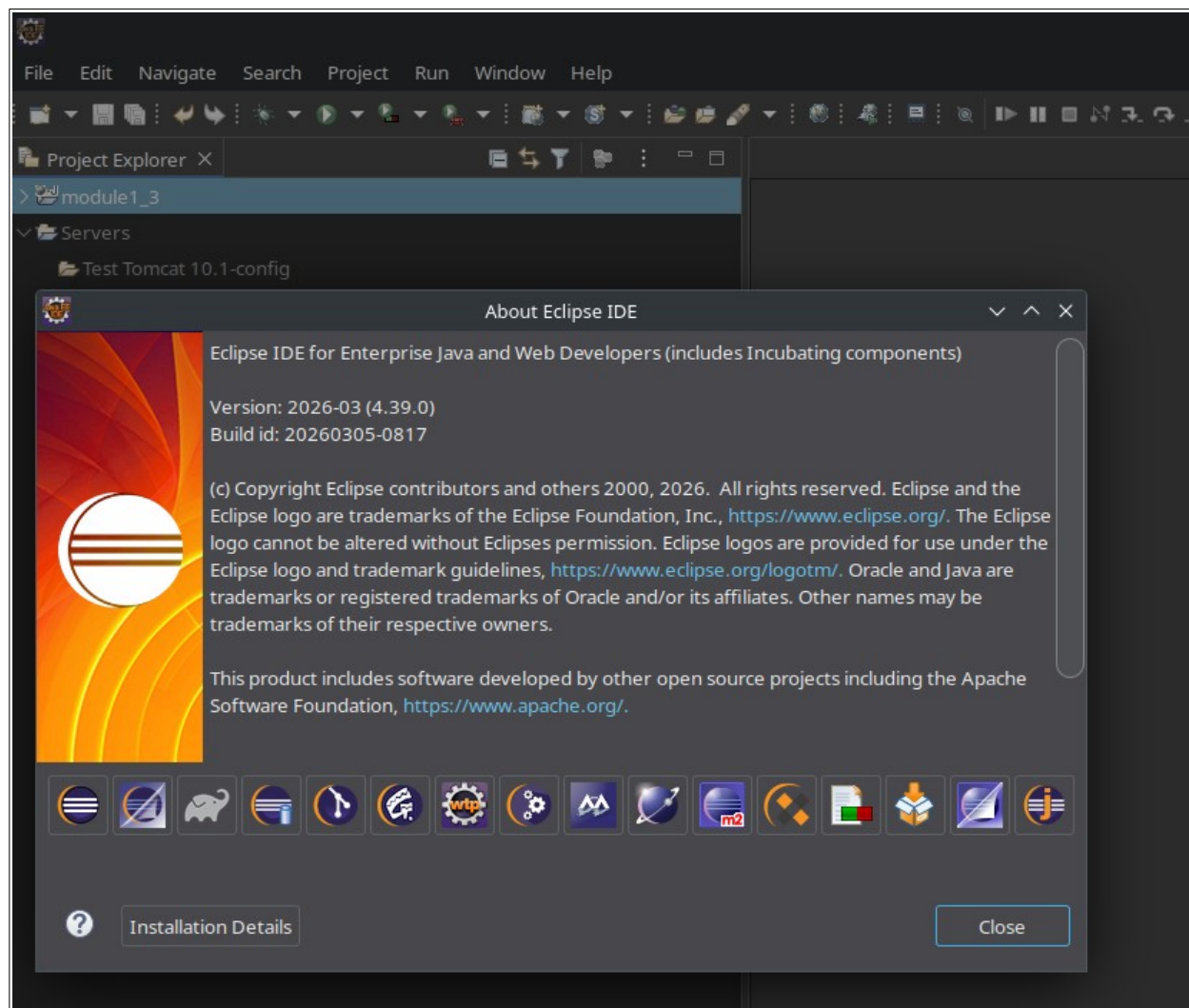


CSD-430: Server Side Development
Module 1: Installation & Configuration
Assignment 3: Programming Assignment
Isaac Ellingson
6/7/2026

Java Installed

```
[falkreon@circe] ~ & java --version
openjdk 25.0.2 2026-01-20
OpenJDK Runtime Environment (build 25.0.2+10-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 25.0.2+10-Ubuntu-124.04, mixed mode, sharing)
[falkreon@circe] ~ & █
```

Eclipse Installed and Configured for Java EE



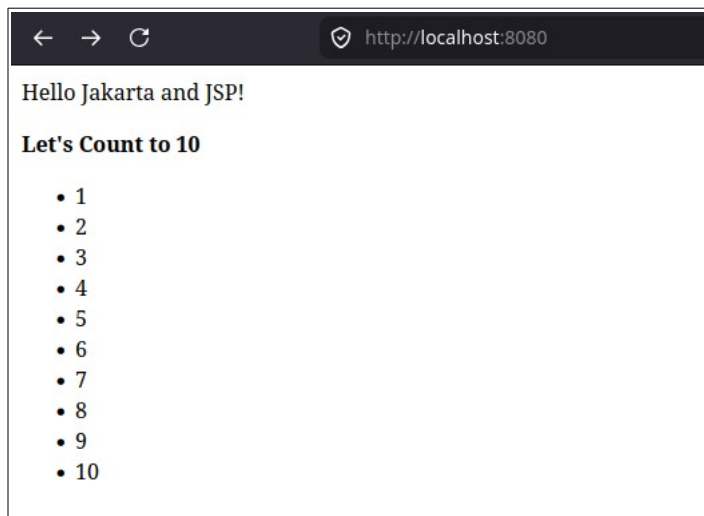
Tomcat Installation

```
[falkreon@circe] ~ & sudo service tomcat10 status
● tomcat10.service - Apache Tomcat 10 Web Application Server
   Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-06-07 19:20:58 CDT; 37min ago
     Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
   Main PID: 315972 (java)
    Tasks: 45 (limit: 76577)
  Memory: 130.3M (peak: 211.6M)
     CPU: 5.202s
    CGroup: /system.slice/tomcat10.service
            └─315972 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/con

Jun 07 19:20:58 Circe tomcat10[315972]: OpenSSL successfully initialized [OpenSSL 3.0.13 30 Jan 2024]
Jun 07 19:20:58 Circe tomcat10[315972]: Initializing ProtocolHandler ["http-nio-8080"]
Jun 07 19:20:58 Circe tomcat10[315972]: Server initialization in [280] milliseconds
Jun 07 19:20:58 Circe tomcat10[315972]: Starting service [Catalina]
Jun 07 19:20:58 Circe tomcat10[315972]: Starting Servlet engine: [Apache Tomcat/10.1.16 (Ubuntu)]
lines 1-16...skipping...
● tomcat10.service - Apache Tomcat 10 Web Application Server
   Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-06-07 19:20:58 CDT; 37min ago
     Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
   Main PID: 315972 (java)
    Tasks: 45 (limit: 76577)
  Memory: 130.3M (peak: 211.6M)
     CPU: 5.202s
    CGroup: /system.slice/tomcat10.service
            └─315972 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/con

Jun 07 19:20:58 Circe tomcat10[315972]: OpenSSL successfully initialized [OpenSSL 3.0.13 30 Jan 2024]
Jun 07 19:20:58 Circe tomcat10[315972]: Initializing ProtocolHandler ["http-nio-8080"]
Jun 07 19:20:58 Circe tomcat10[315972]: Server initialization in [280] milliseconds
Jun 07 19:20:58 Circe tomcat10[315972]: Starting service [Catalina]
Jun 07 19:20:58 Circe tomcat10[315972]: Starting Servlet engine: [Apache Tomcat/10.1.16 (Ubuntu)]
Jun 07 19:20:58 Circe tomcat10[315972]: Deploying web application directory [/var/lib/tomcat10/webapps/ROOT]
lines 1-17...skipping...
● tomcat10.service - Apache Tomcat 10 Web Application Server
```

Simple JSP Page



JSP Page Source (index.jsp)

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Test JSP Page</title>
</head>
<body>
    <%
String HELLO = "Hello Jakarta and JSP!";
out.println(HELLO);
    %>
<p><b>Let's Count to 10</b>
<ul>
    <%
for(int i=1; i<=10; i++) {
    out.println("<li>" + i);
}
    %>
</ul>
</body>
</html>
```